

Tesfaye Sebastiano Salarin

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Work Experience

Financial Engineer - Quantitative Analyst - Mediobanca, Milan (Italy) 03/2026 – Present
Financial Engineering Division

- **Developed Skills:** Derivatives Pricing, Fixed-Income Market Making, Market Impact Modelling, Large Language Models (LLMs), Vector Embeddings and Vector Databases, Model Benchmarking, Token Usage and Cost Optimisation, Sentiment Analysis, and Software Infrastructure Engineering

TA in *Strategie d'Investimento e derivati*, MSc in Economia e Finanza 09/2025 – 12/2025
University of Trento, Department of Economics & Management

- **Developed Skills:** Modern Portfolio Theory (e.g., Single Index Model, Markowitz, Black-Litterman), Portfolio Optimization, Risk measurement, Time Series Analysis, MATLAB Coding, Copilot and AI for Finance

TA in *Data Analysis and Forecasting*, MSc in International Management e LM in Innovation Management Course 09/2025 – 12/2025
University of Trento, Department of Economics & Management

- **Developed Skills:** Time Series graphics and decomposition, Time Series features, Regression Analysis, Exponential Smoothing

TA in *Introduction to Computer Programming*, BSc in Economics and Management 09/2022 – 04/2024
University of Trento, Department of Economics & Management

- **Developed Skills:** Data collection, Data Analysis, Data Cleaning, Data Wrangling/Transformation (tidyverse package), Data Visualization RMarkdown, RShinyApp, Debugging/Troubleshooting, Project & Time Management, Effective Teaching Techniques (i.e., Collaboration/Communication)

TA in *Microeconomics*, BSc in Economics and Management 09/2022 – 12/2023
University of Trento, Department of Economics & Management

- **Developed Skills:** Consumer Theory, General Equilibrium, Welfare Economics, Oligopoly, Game Theory, economic reasoning, problem-solving, inter-communicative skills, time management, class schedules, assessment and marking.

TA in *Computational Finance*, BSc in Economics, Markets and Finance 01/2021 – 03/2021
Cà Foscari University of Venice, Department of Economics

- **Developed Skills:** Monte-Carlo Simulations, RStudio, quantitative finance, data analysis, problem-solving, inter-communicative skills, time management, class schedules, assessment and marking

Akos S.r.l. 02/2021 – 09/2021
 Internship - Junior Analyst

- **Developed Skills:** Assisted in M&A deals (from beginning to Due Diligence), company valuation models (DCF, multiples) for SMEs under management

Capone-Salmeri-Vianello Dott.ri Commercialisti Ass. 05/2018 – 08/2018
 Internship – Accounting Assistant

- **Developed skills:** Assisted in accounting tasks, deals negotiation, and selling site-visits

Education

EconopysiX Lab - Capital Fund Management (CFM), Paris (France) 09/2025 – 02/2026
Visiting Ph.D. at the [EconopysiX Lab - CFM Chair of Econophysics & Complex Systems at Ecole Polytechnique](#) [🔗](#) Research Group

University College of London, Department of Computer Science, London (UK)

09/2024 – 03/2025

Visiting Ph.D. at the [Financial Computing & Analytics](#)  Research Group

University of Trento, Department of Economics & Management

10/2021 – Present

Ph.D. in Economics and Management

- **Research Interests** : Market Microstructure, High-Frequency Trading, Algorithmic Trading, FinTech, De-Fi, Econometrics, NLP, Machine-Learning and AI for financial application

Cà Foscari University of Venice, Department of Economics

09/2019 – 10/2021

MS in Finance, with honors - graduated *Cum Laude*

- **Courses**: Derivatives and Insurance, Computational Finance Lab, Advanced Econometrics (in Gretl and RStudio), Stochastic Calculus for Finance (e.g., solving ODEs, PDEs, BSM Model, pricing derivatives), Stochastic Models for Finance (e.g., Time-Series analysis, ARIMA models, Regressions), Statistical Methods for Risk Analysis (e.g., VaR, ES, multivariate distributions), Risk Management (MATLAB), Financial Reporting, Corporate Finance.

Publications and Projects

Research Papers

- Estimating the short-term stock market reaction to non-financial information: an Event-Study approach on ESG News (2024) - [SSRN](#) 
- Cryptocurrencies in the Balance Sheet: Insights from (Micro)Strategy – Bitcoin Interactions(2025) - [arXiv](#) 

Other projects

- Co-supervised a BSc Thesis on Modern Portfolio Theory (i.e., Portfolio optimization with Black-Litterman and Markowitz models) using Microsoft Excel and RStudio - (2025)
- Co-supervised a BSc Thesis on The Impact of the 2009, 2012, and 2016 Earthquakes in Italy on Firm-Level Returns (Event-Study Methodology) using Microsoft Excel RStudio, and Python programming language (2025)

Awards

VERA Academy Internship Grants - Research Assistant, University of Cà Foscari

- At VERA Academy from March 2021 to October 2021, I collected data on local public finances of Italian municipalities from 1998-2018, carried out data-cleaning and econometric analysis for a report to be included in the *Ca' Foscari Report on Italian Municipalities*

Summer Schools

- **SiDE Postgraduate Course on Financial Time Series and High-Frequency Econometrics**
S.A.Di.Ba. – Scuola di Automazione per Dirigenti Bancari, Banca d'Italia, Perugia (Italy) - [SiDE Summer School](#) 
- **Summer School in Market Microstructure**
(led by Prof. Björn Hagströmer, Prof. Albert J. Menkveld & Prof. Thierry Foucault) - [Microstructure Course](#) 

Skills

Programming Languages: R (Advanced), Python (Intermediate), SQL (Intermediate, Stata (Intermediate), MATLAB (Beginner), C++/C# (Beginner)

AI and Data Engineering: LLMs (cloud/local integration, Agentic Workflow (LangChain/LangGraph), Natural Language Processing, Sentiment Analysis, Vector Embeddings, Vector Databases, Model Evaluation and Comparison, Token Management and Cost Optimisation, Retrieval-Augmented Generation (RAG)

Databases and Infrastructure: OracleDB, ChromaDB, MongoDB, Software Infrastructure

Research and Productivity Tools: Office 365, Git, L^AT_EX

Financial Data Platforms: LSEG Refinitiv Eikon (Intermediate), Bloomberg Terminal (Beginner), LOBSTER High-Frequency Data

Spoken Languages: Italian (Native language), English (Advanced - B2, FCE), French (Intermediate - B2, Baccalauréat, Double High-School Degree), German (Intermediate - B1, Goethe Institut), Spanish (Beginner)

Conferences, Workshops & Talks

Conferences (Presenter): MAF2024, EWGCFM 69th, FBA Conference, Global Finance Conference, 5th PhD Showcase (UCL).

Workshops: Generative AI in Finance Workshop, Algo-Trading & DeFi Workshop.

Talks: Blockchain Technology and Sustainability (UCL).

Acconsento alla pubblicazione del mio CV in ottemperanza alle disposizioni di legge dettate in materia di trasparenza (D.Lgs. 33/2013).

I consent to the publication of my CV in compliance with the legal provisions on transparency (Legislative Decree 33/2013).